RE07 Global Infrastructure Investment and Management

2. Institutional Frameworks and Stakeholder Governance

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Lecture Outline

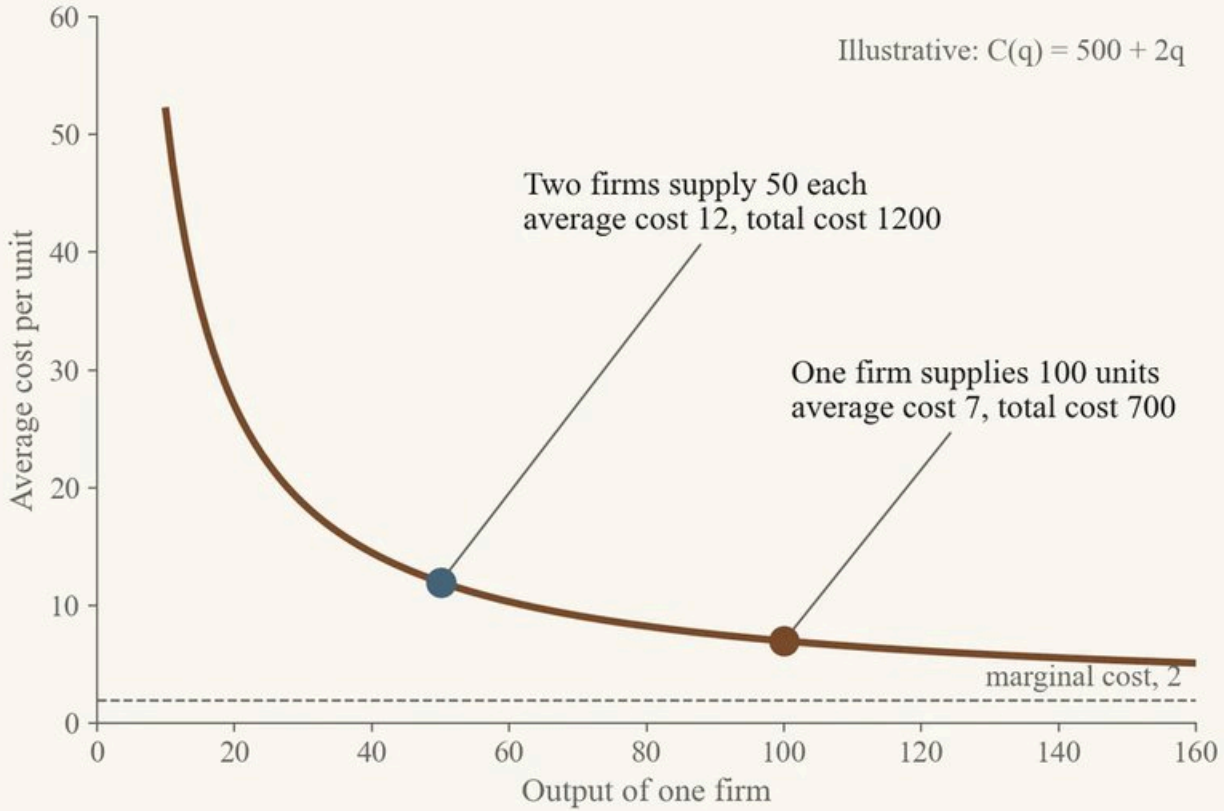
The role of the state
Public investment management
Regulation of natural monopoly
Delivery models and the PPP framework
Development finance institutions
Concessions and renegotiation
Case Study: Heathrow Airport Expansion (Hannah Newell, DfT)

The Role of the State

Problem	Why it arises	What the state does
Natural monopoly	One firm supplies the market at lower total cost than two	Owns, franchises or regulates the supplier
Sunk costs	Assets cannot be moved, so the firm can be held up	Gives legal protection and credible commitments
Incomplete contracts	Contracts cannot specify quality in full	Chooses who owns, builds and operates
Asymmetric information	The firm knows its costs better than the regulator	Audits, benchmarks and sets incentives

Joskow, P. L. (2005), Regulation of Natural Monopolies, MIT CEEPR Working Paper 05-008, pages 8, 20–21, 93–94 and 119; Shleifer, A. (1998), State versus private ownership, Journal of Economic Perspectives 12(4), pages 8–9 of the NBER version.

Natural Monopoly: Subadditive Costs



Definition: one firm supplies the market at lower total cost than two or more

Scale economies: they are sufficient, not necessary: demand may be too small for two efficient firms

Sunk costs: they explain why a single supplier survives, and why it can be held up after it invests

The trade: a monopoly can be cheaper to run and costlier to use

Joskow, P. L. (2005), Regulation of Natural Monopolies, MIT CEEPR Working Paper 05-008, pages 8–11 and 20–21. Cost function C(q) = F + cq as in Joskow, page 9; parameters illustrative.

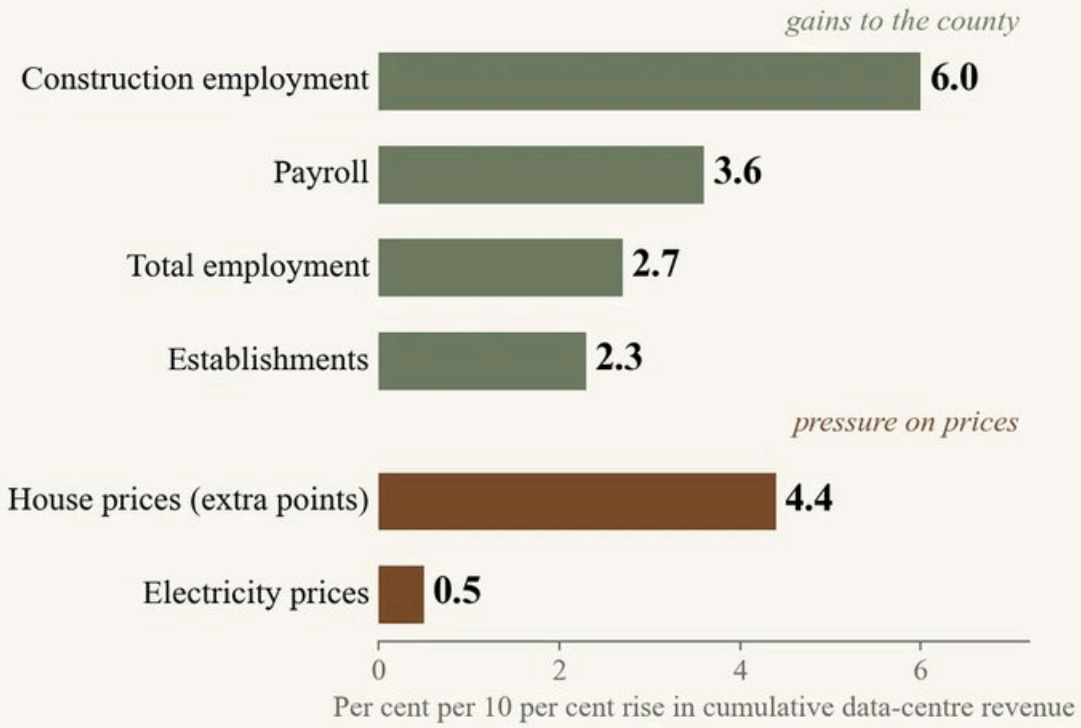
Public Ownership, Franchising or Regulation

Response	Idea	Main weakness
Public ownership	The government owns and runs the supplier	Weak incentives to cut cost and to innovate
Franchise bidding	Competition for the market replaces regulation	Needs many independent bidders; the contract is incomplete
Regulation	A regulator sets prices that cover cost	The regulator knows less than the firm

Where cost cutting harms quality that cannot be written into a contract, public ownership can be the better arrangement (Shleifer, 1998, p. 11).

Shleifer, A. (1998), State versus private ownership, Journal of Economic Perspectives 12(4), pages 8–11 of the NBER version; Joskow, P. L. (2005), MIT CEEPR Working Paper 05-008, pages 64–66 and 93–94.

Data Centres and the Local Economy



Method: an instrument built from nineteenth-century railway corridors and fibre access points

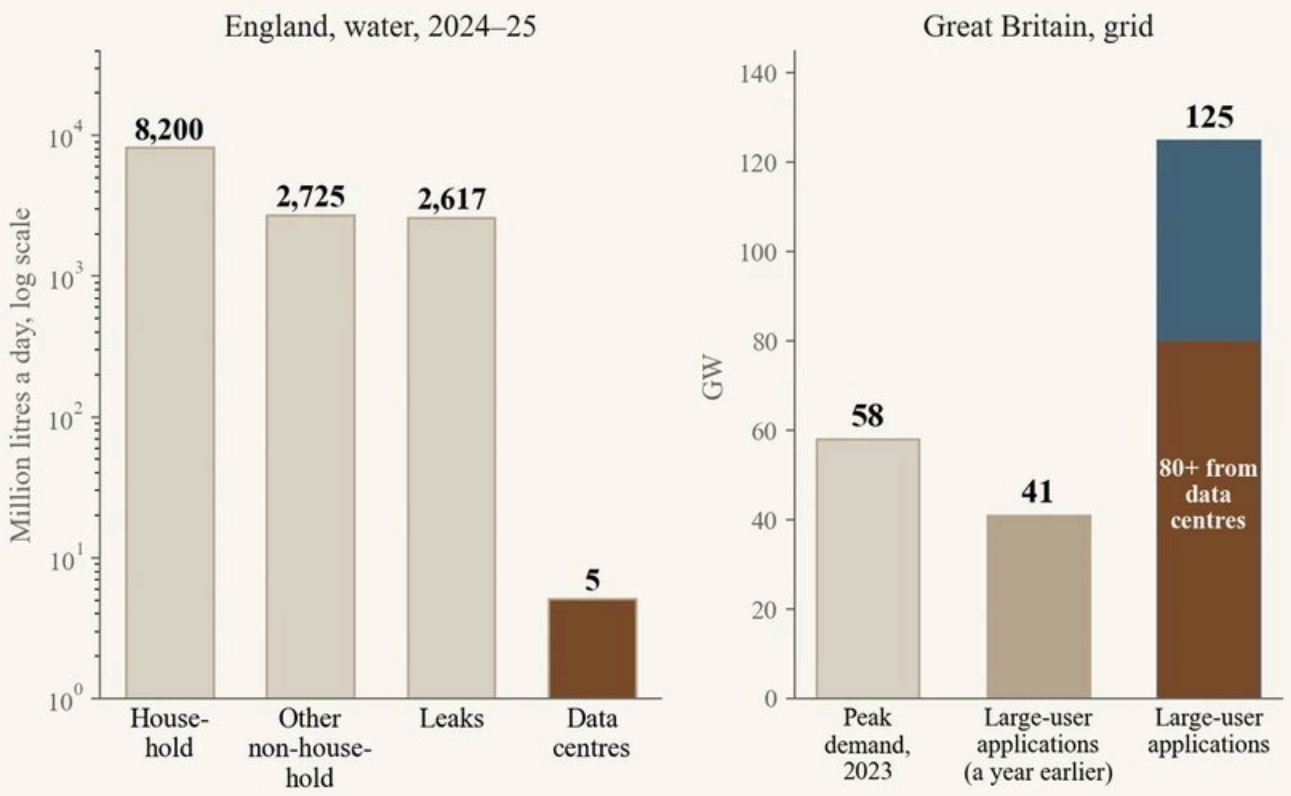
Gains: employment, payroll and the number of firms rise in the host county

Pressures: house prices, electricity prices and public water withdrawals rise as well, the last two later

Wholesale: data centres raised US wholesale power prices by 6.9 per cent by 2025 and by 15.0 per cent in northern Virginia

Alvarez, F. E., Argente, D., Chow, J. and Van Patten, D. (2026), NBER Working Paper 35194, pages 2 to 4; Kay, O., Reaser, R. and Taylor, R. (2026), Federal Reserve Bank of Dallas Working Paper 2606, abstract.

National Totals and Local Constraints: Water and the Grid



Water: data centres use 5 million litres a day in England, a five-hundredth of what leaks lose

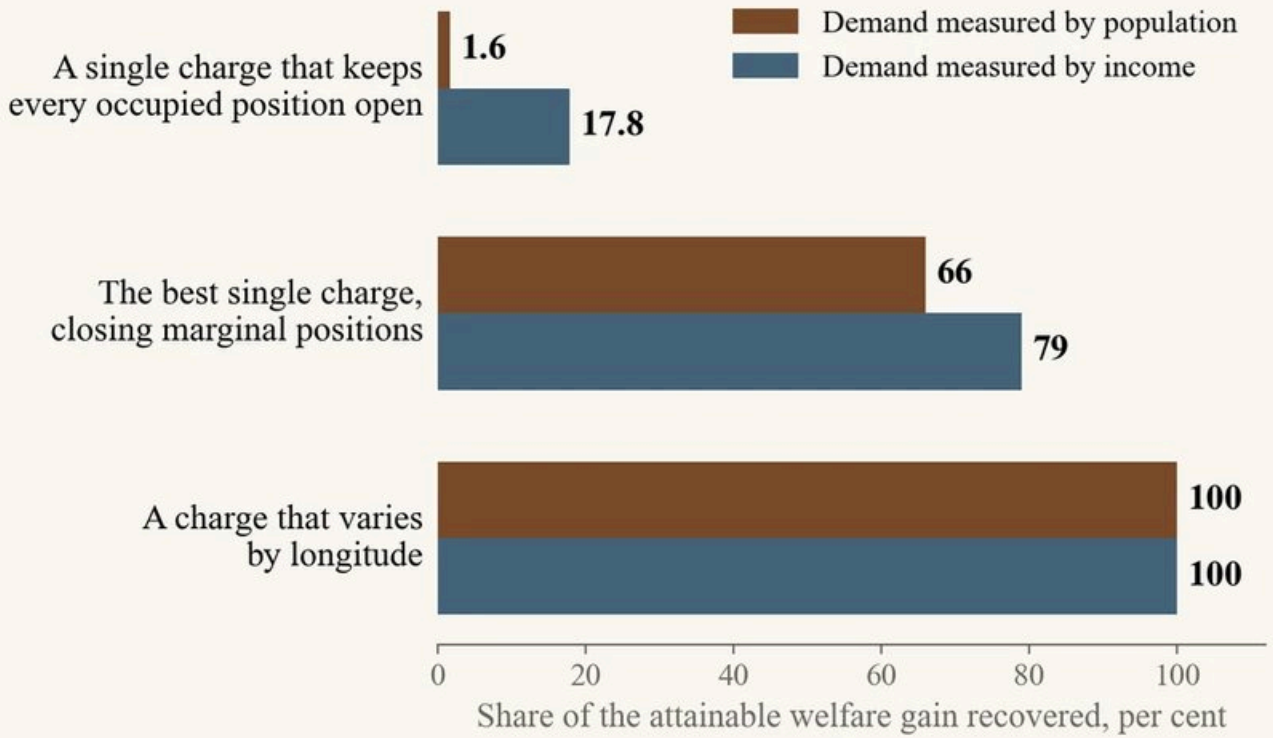
Concentration: the six largest of 257 surveyed use 65 per cent of the sector's water

Grid: applications from large users reached 125 GW against a peak demand of 58 GW, at least 80 GW from data centres

Siting: water use per kilowatt-hour varies by place and hour, so location decides the effect

Environment Agency (2024 to 2025); WRc and MOSL (2026), Water Efficient Data Centres; Ofgem figures as reported in the Spectator's Reality Check (August 2026); Li, P., Yang, J., Islam, M. A. and Ren, S. (2025), Making AI less thirsty, Communications of the ACM, DOI 10.1145/3724499.

A Commons of Unequal Locations: The Geostationary Arc



The commons: no state may own a position; the ITU records filings but sets no price and no cap

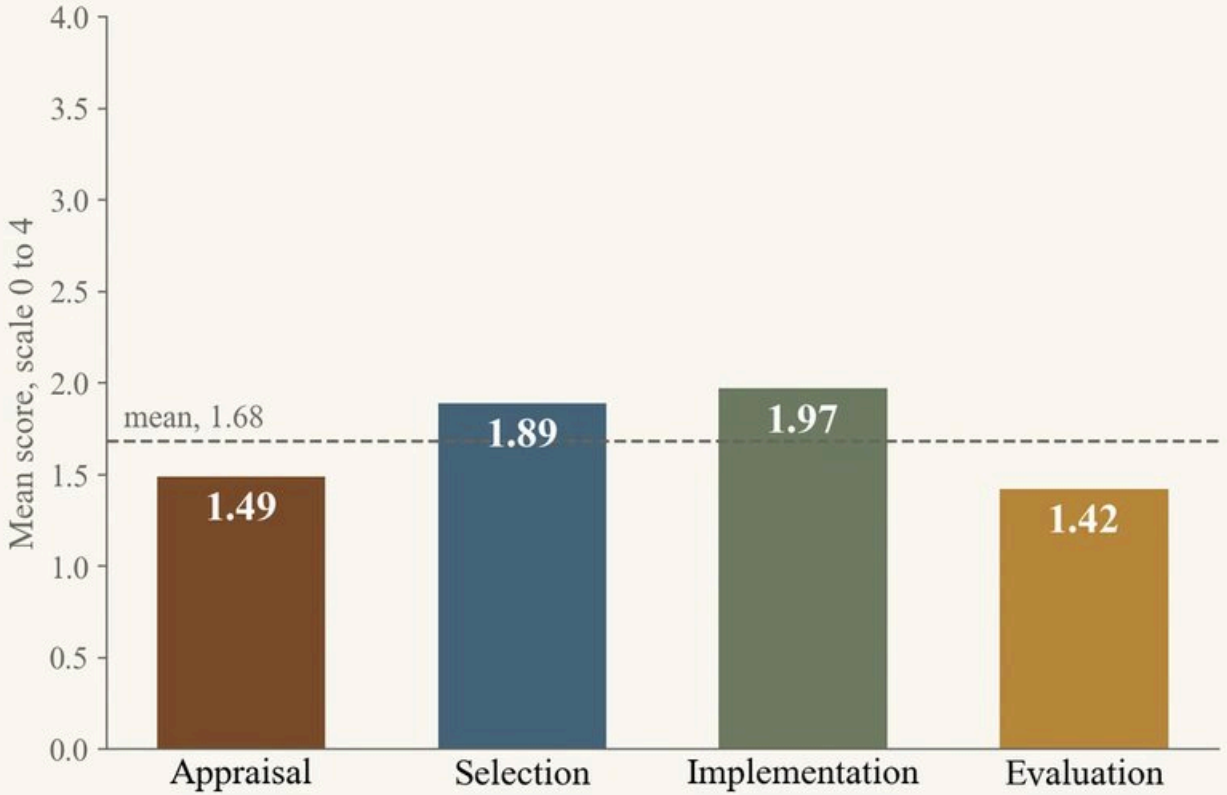
The error: free entry equalises average, not marginal, cost, so each position admits too many satellites

The price: the efficient charge differs by a factor of 748 between the busiest and thinnest positions

One charge: it recovers 66 per cent of the attainable gain only by closing 27 of 65 occupied bins of five degrees

Wagner, A. F. (2026), The Price of Position: Spatial Equilibrium in the Geostationary Arc, SSRN 7576078 (working paper, 25 October 2026 version), pages 1 to 5. Bars: shares of the attainable welfare gain; model estimates.

Public Investment Management



Four stages: appraisal, selection, implementation and evaluation, scored from 0 to 4 for 71 countries

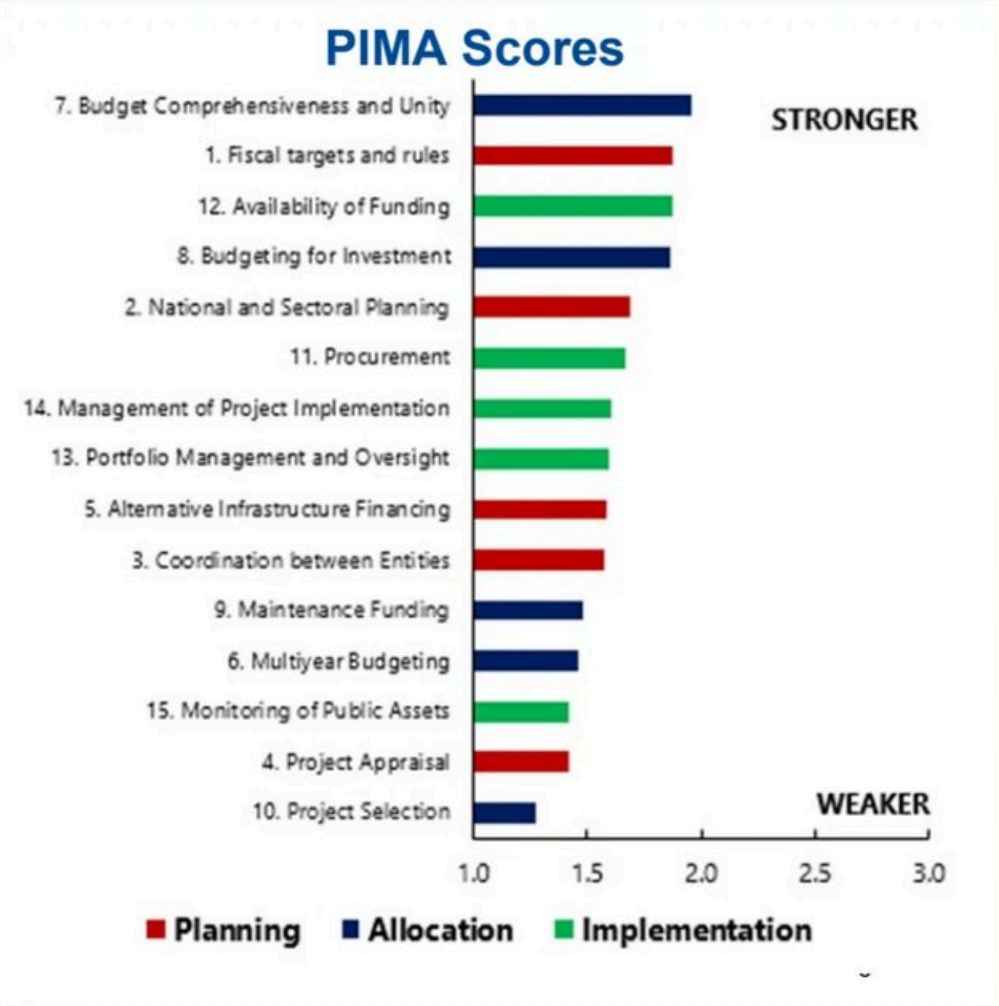
Weak points: appraisal and evaluation score lowest in every region

Spread: scores run from South Africa and Colombia at the top to Belize and Yemen at the bottom

Income: Bolivia and Rwanda score above 2, ahead of 19 middle-income countries

Dabla-Norris, E., Brumby, J., Kyobe, A., Mills, Z. and Papageorgiou, C. (2011), Investing in public investment: an index of public investment efficiency, IMF Working Paper 11/37, pages 14–17.

PIMA: Design Outruns Effectiveness



PIMA: scores 15 institutions across planning, allocation and implementation

Design and practice: institutions are generally stronger on paper than in implementation

Effectiveness: it is low at every stage and in every income group

The gain: better governance could make up more than half of observed efficiency losses

Verdier, G. (2020), Strong Infrastructure Governance for Strong Recovery, IMF Fiscal Affairs Department, slides 4 and 5 (CC0); Schwartz, G., Fouad, M., Hansen, T. and Verdier, G. (eds) (2020), Well Spent, IMF.

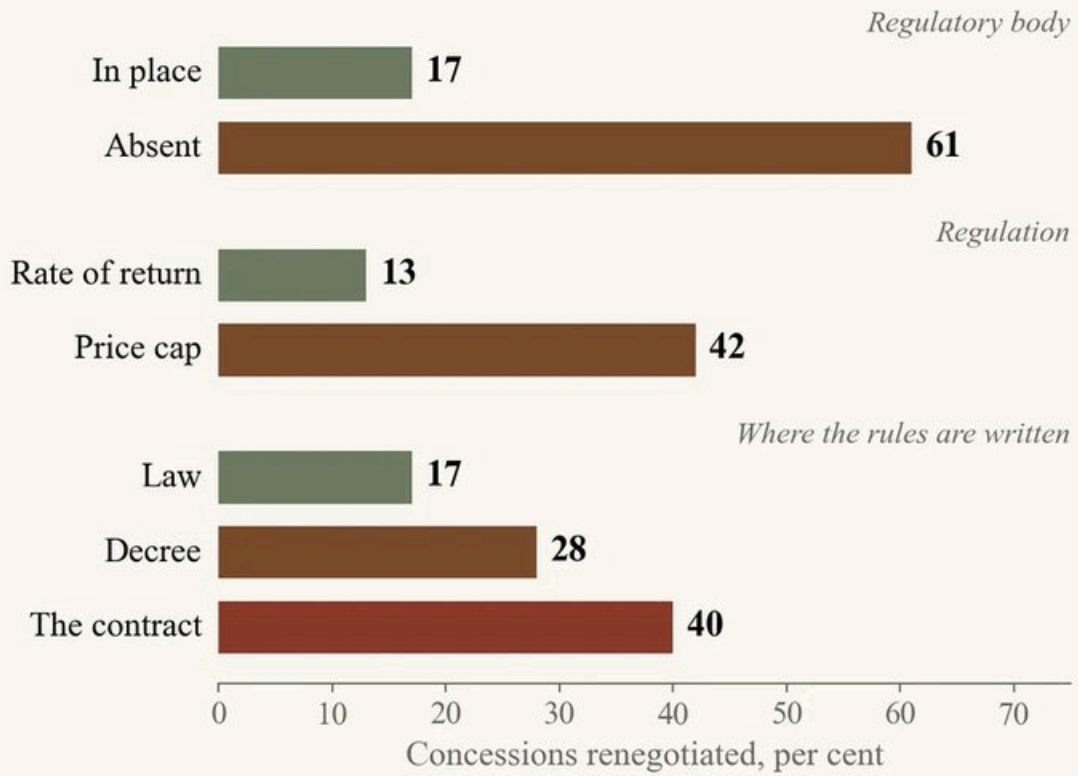
Cost-of-Service Regulation and Price Caps

	Cost-of-service regulation	Price cap or sliding scale
How price is set	Operating cost, depreciation and a return on the regulatory asset base, reset in a rate case	A price path fixed in advance, partly responsive to realised cost
What the firm keeps	Savings until the next rate case	Savings between reviews, shared by formula
Main risk	Too much capital, since the return is on the base	Cost cutting that lowers quality; ratchet effects

Revenue requirement = operating cost + depreciation + allowed return × regulatory asset base + other costs (Joskow, 2005, p. 98).

Joskow, P. L. (2005), Regulation of Natural Monopolies, MIT CEEPR Working Paper 05-008, pages 94–99, 113–115, 119–128 and 146.

Regulatory Design and Renegotiation



Sample: more than 1,000 concessions in Latin America and the Caribbean, 1985 to 2000

A regulator: 17 per cent renegotiated where one existed, 61 where none did

The regime: price caps 42 per cent, rate of return 13

The rules: 17 per cent when set in law, 28 in a decree and 40 in the contract

Guasch, J. L. (2004), Granting and Renegotiating Infrastructure Concessions: Doing It Right, World Bank Institute, Table 1.16 (page 17) and pages 12 and 16.

Capture, Ratchets and Hold-up

Problem	Mechanism	Institutional response
Information gap	The firm knows its costs and effort better than the regulator	Uniform accounts, audits, benchmarking
Capture	The firm or another group bends the regulator's decisions	Transparency, appeals, legislative oversight
Ratchet	Savings revealed now are taken at the next review	Regulatory lag; profit sharing
Hold-up	Sunk assets invite expropriation after investment	Legal protection; credible commitment

Joskow, P. L. (2005), Regulation of Natural Monopolies, MIT CEEPR Working Paper 05-008, pages 61, 93–99, 119–120, 126 and 146.

Delivery Models and the PPP Framework

The delivery spectrum: from management contracts to privatisation, and who bears which risk

Bundling build and operate: Hart (2003): what the contract can specify decides the choice

Sharing demand risk: Engel, Fischer and Galetovic (2013): a guarantee, a cap and a variable term

The price of private finance: National Audit Office (2018): what the bundle must save

The framework around the contract: Verhoest and co-authors (2015): government support across 20 countries

The Delivery Spectrum

	Management contract	Concession	Privatisation
Ownership of assets	Government	Government	Private operator
Investment	Government	Private operator	Private operator
Service control	Government	Shared	Shared
Tariff control	Government	Shared	Shared
Revenue risk	Government	Private operator	Private operator
Cost risk	Government	Private operator	Private operator
Labour risk	Government	Private operator	Private operator
Management cost risk	Private operator	Private operator	Private operator

Between public supply and private supply lie outsourcing, leases, franchises, build-operate-transfer and divestiture.

Guasch, J. L. (2004), Granting and Renegotiating Infrastructure Concessions: Doing It Right, World Bank Institute, Table 2.1 (page 25) and Figure 2.1 (page 24).

Hart (2003): Bundling Build and Operate

Question: should one firm build and run the asset, or should separate firms?

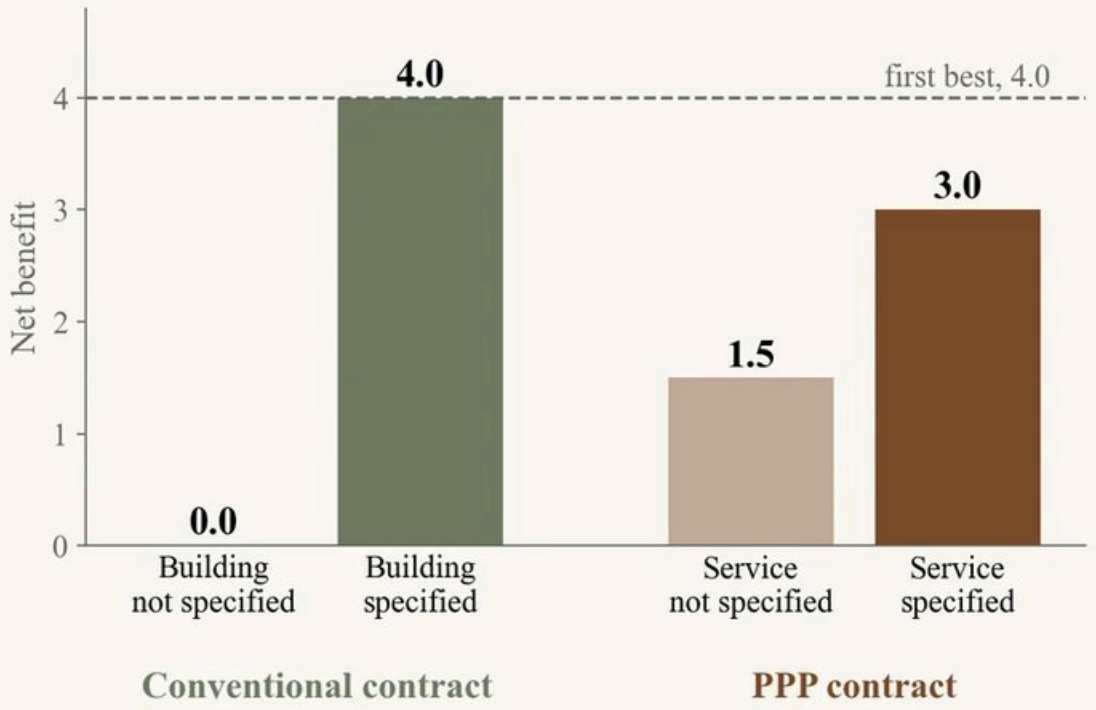
Model: the builder makes two investments the contract cannot verify: a better building, and corner-cutting that lowers cost and quality

Result: bundling gives the builder the operating saving, so he invests more in the building and cuts more corners

Rule: conventional provision if the building can be specified and the service cannot; a PPP in the opposite case

Hart, O. (2003), Incomplete contracts and public ownership: remarks, and an application to public-private partnerships, Economic Journal 113(486), C69–C76; CMPO Working Paper 03/061, pages 5–11.

Bundling and the Limits of the Contract



Neither specified: the PPP beats conventional provision, 1.5 against 0, because the builder invests more

Service specified: the PPP rises to 3 as corner-cutting is penalised

Building specified: conventional provision reaches the first best, 4

Author's calculation from Hart's (2003) model, with functional forms chosen by the author (models_w2.py); psi = 1, beta = 1, gamma = 2, delta = 1.5. CMPO Working Paper 03/061, pages 6–10.

Engel, Fischer and Galetovic (2013): Sharing Demand Risk

Setting: a risk-averse firm builds and runs a project whose demand is uncertain, and subsidies waste more than user fees

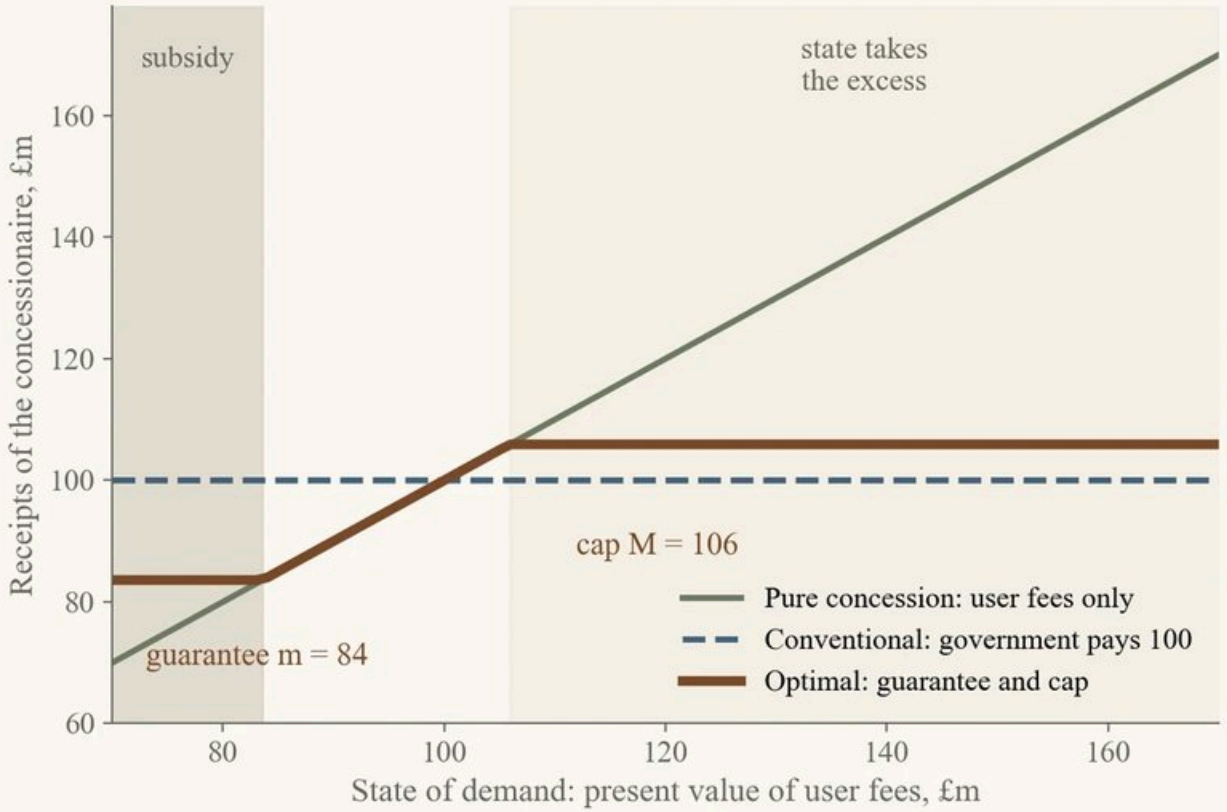
First result: PPPs do not release public funds: the saving now is given back as fees forgone later

Optimal contract: a minimum revenue guarantee, a revenue cap and a variable term

In practice: observed guarantees and revenue-sharing clauses bear no relation to the optimum

Engel, E., Fischer, R. and Galetovic, A. (2013), The basic public finance of public-private partnerships, Journal of the European Economic Association 11(1), 83–111; NBER Working Paper 13284, pages 2–6, 12–16 and 41.

The Optimal Concession Contract



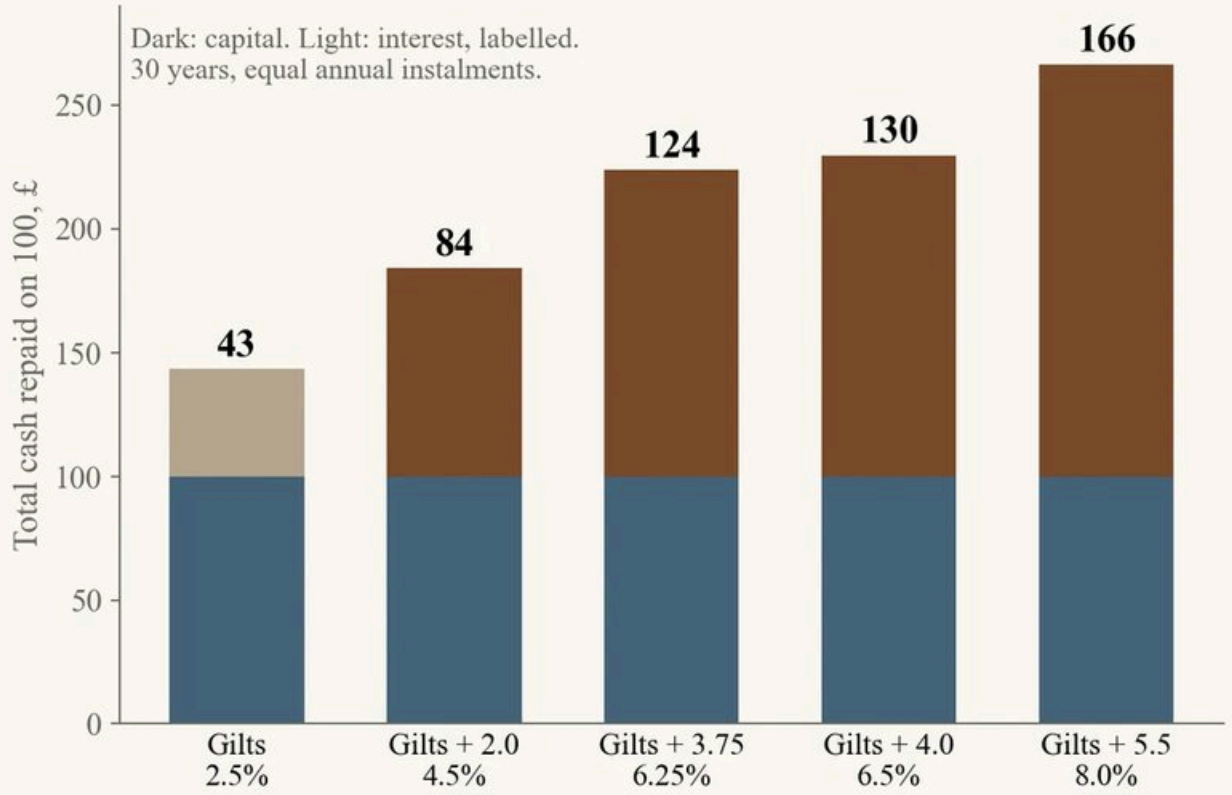
Low demand: the concession runs on and the state tops up to the guarantee

Intermediate: the firm keeps what it collects, and no subsidy is paid

High demand: the concession ends at the cap, and the state takes the excess

Author's calculation from the model of Engel, Fischer and Galetovic (2013), NBER Working Paper 13284, pages 12–16, with exponential utility and uniform demand chosen by the author; I = 100, demand 70–170, risk aversion 0.01, zeta = 0.25, lambda = 0.3.

Private Finance and the Public Sector Comparator



The premium: 2 to 3.75 points over gilts in the 2010 plan, 2 to 4 for deals since 2013, and over 8 per cent in four 2013 housing deals

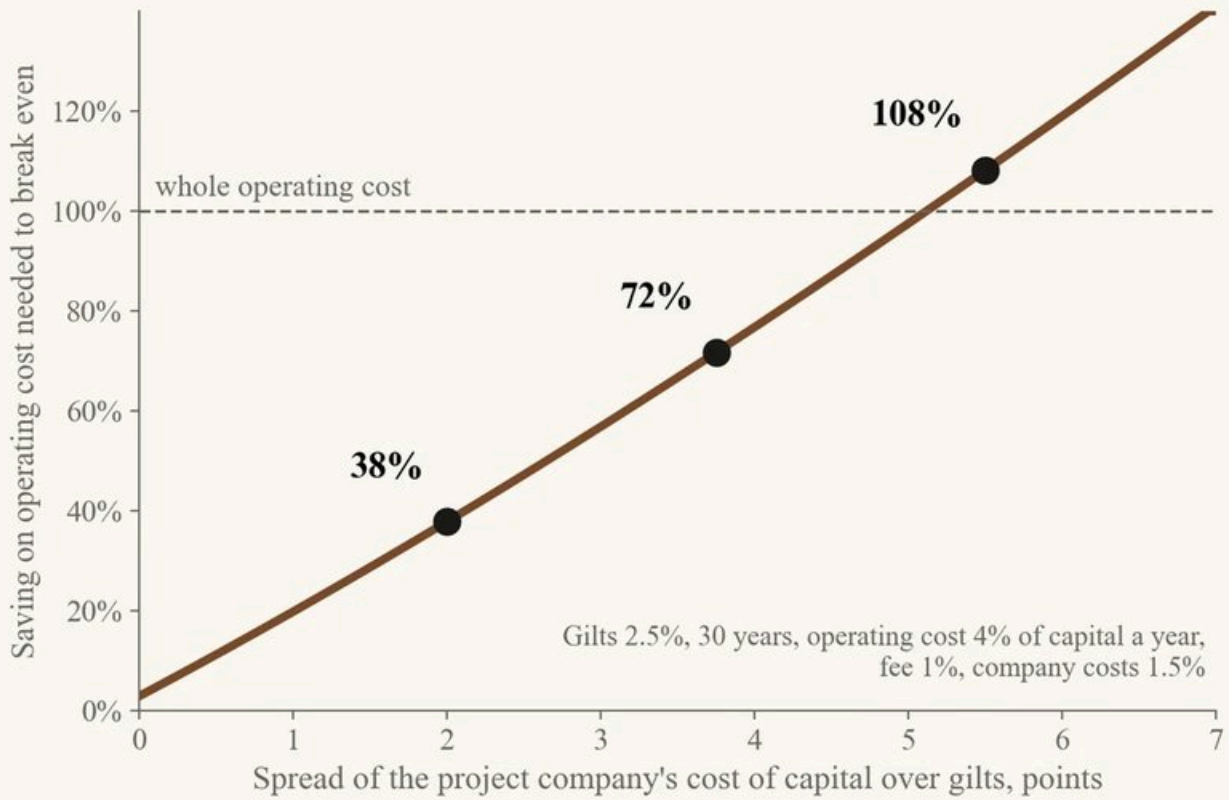
Compounding: £100 over 30 years costs £34 in interest at 2 per cent and £73 at 4

Comparators: PF2 schools cost about 40 per cent more than government borrowing, a privately financed hospital 70 per cent more

The offset: no evidence of operational efficiency in the NAO's hospital samples

National Audit Office (2018), PFI and PF2, HC 718, paragraphs 1.9 and 1.17–1.19 and Figures 3 and 4 (pages 10 and 14–16); bars computed by the author from the annuity formula, which reproduces Figure 3.

The Saving Needed to Offset Private Finance



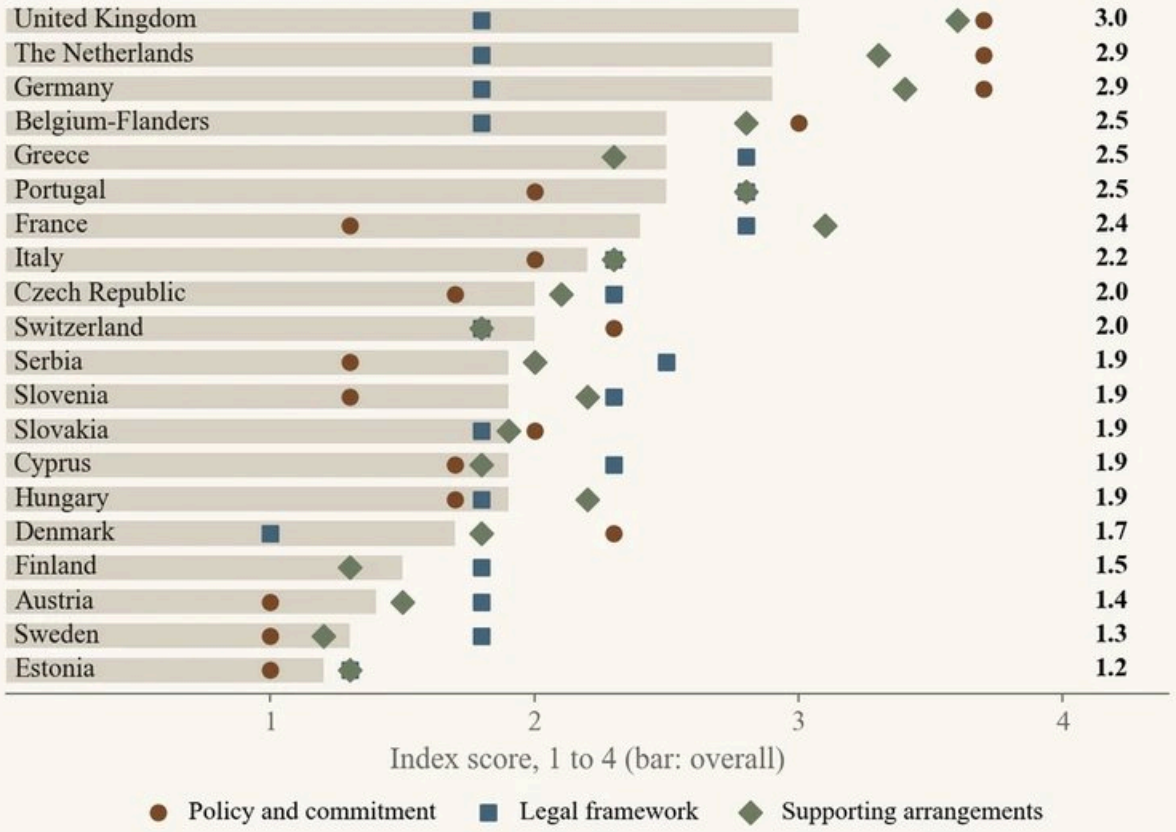
Gilts plus 2.0: the bundle must cut operating cost by 38 per cent

Gilts plus 3.75: 72 per cent

Gilts plus 5.5: more than all of it, 108 per cent

Author's calculation after National Audit Office (2018), PFI and PF2, paragraphs 1.17–1.19, 30 years, gilts 2.5%, operating cost 4% of capital a year (author's assumption), fee 1%, company costs 1.5%.

PPP Frameworks across Europe



Index: policy commitment, legal framework and supporting arrangements in 20 countries, each scored from 1 to 4

Range: the United Kingdom scores 3.0, the highest, and Estonia 1.2, the lowest

No package: the UK, the Netherlands and Germany score high on policy and low on PPP law

Uptake: a correlation of 0.31 with the value of PPPs relative to GDP, and 0.58 with the number of projects

Verhoest, K., Petersen, O. H., Scherrer, W. and Soecipto, R. M. (2015), Transport Reviews 35(2), 118–139; Murwantara, R. (2023), doctoral thesis, University of Antwerp, chapter 2, Table 3 (page 30) and page 34.

Three Models of the Contract Compared

Model	Explains	Prediction	Evidence
Hart (2003)	Whether to bundle build and operate	Bundle if the service can be specified and the building cannot	Model only; sector examples
Engel et al. (2013)	How to share demand risk	A guarantee and a cap; no budget relief	Renegotiation follows low demand
National Audit Office (2018)	What the bundle must save	The finance premium has to be earned back	Premium of 2–4 points; thin evidence of savings
Verhoest et al. (2015)	What governments do to support PPPs	Support varies and is not a package	Weak link to uptake, r = 0.31

Hart (2003), pages 5–11; Engel, Fischer and Galetovic (2013), pages 2–6 and 15; National Audit Office (2018), paragraphs 1.9 and 1.17–1.19; Verhoest et al. (2015) as reported in Murwantara (2023), chapter 2, page 34.

Development Finance Institutions

Role: multilateral banks lend directly, and are expected to mobilise private lenders

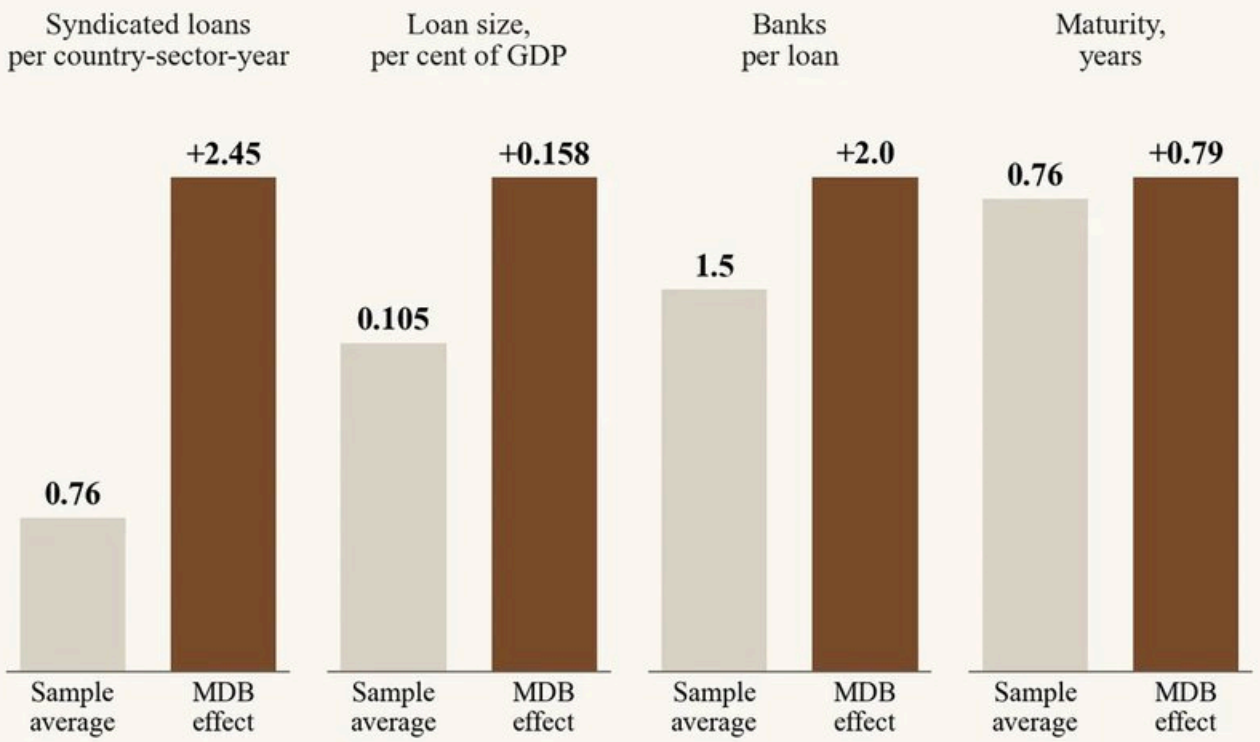
Signalling: lending on commercial terms signals opportunity and can encourage reform

Risk mitigation: preferred creditor status, close monitoring and information reduce political risk for private lenders

Scale: an annual investment gap of about USD 2.5 trillion against aid of USD 143 billion in 2016

Broccolini, C., Lotti, G., Maffioli, A., Presbitero, A. and Stucchi, R. (2019), Mobilization Effects of Multilateral Development Banks, IDB Invest Technical Note 16, pages 1 and 14.

Mobilisation Effects of Multilateral Lenders



Data: syndicated loans to developing countries, 1993 to 2017

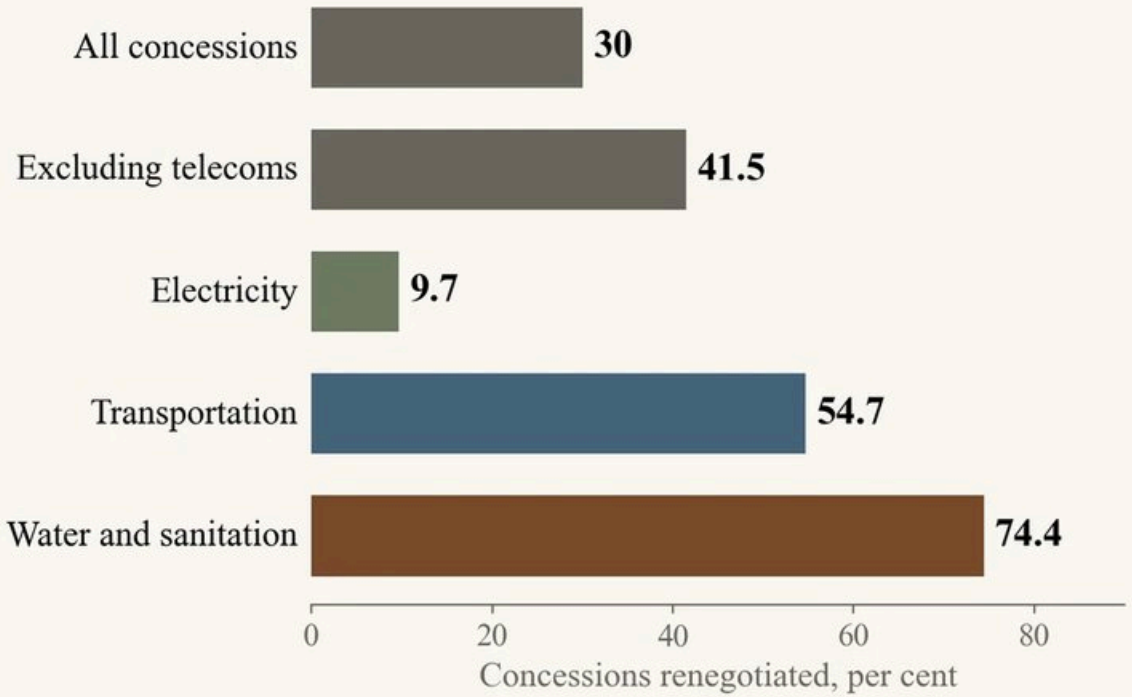
Finding: a multilateral lender present for three years raises the number of loans, their size, the banks per loan and their maturity

Robustness: no decline in bond finance and no sign of anticipation

Caution: the effect on the number of loans is significant only at 10 per cent

Broccolini, C., Lotti, G., Maffioli, A., Presbitero, A. and Stucchi, R. (2019), Mobilization Effects of Multilateral Development Banks, IDB Invest Technical Note 16, Table 3 (page 30).

Concessions and Renegotiation



Frequency: 30 per cent of more than 1,000 concessions were renegotiated, and 41.5 per cent excluding telecoms

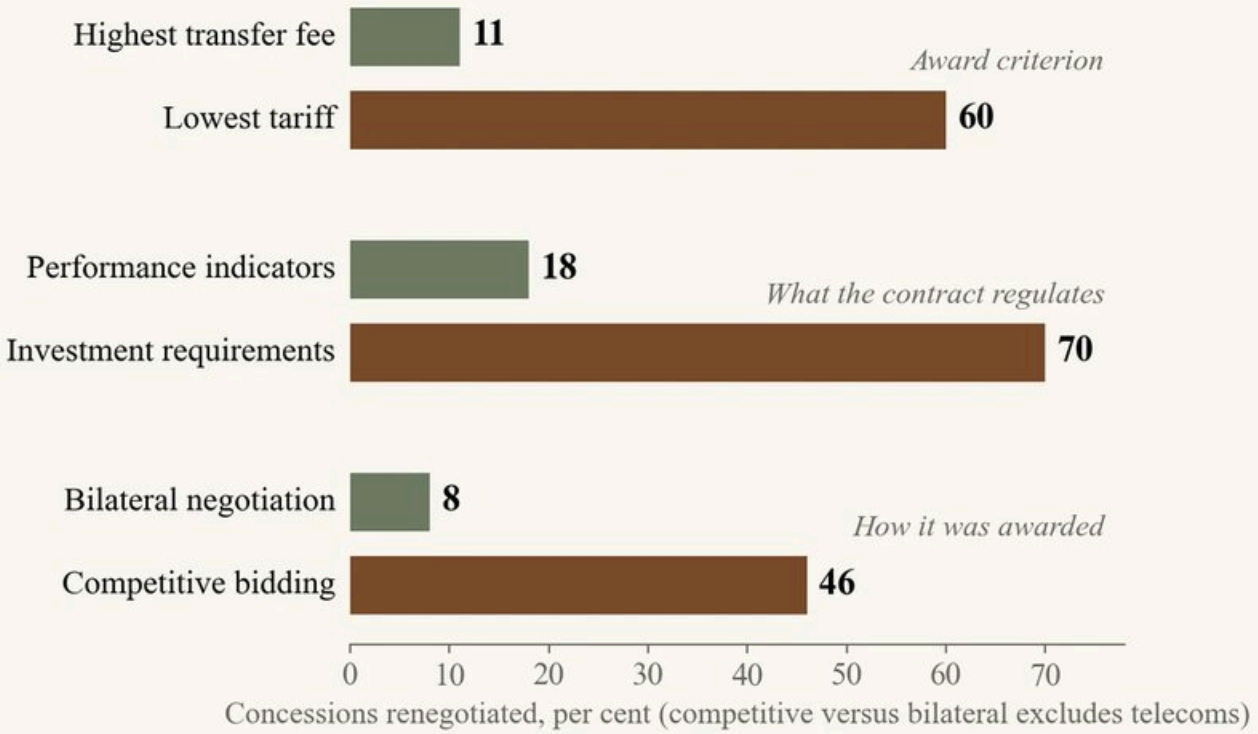
Sectors: water 74 per cent, transport 55, electricity 10

Timing: on average 2.2 years after award, and 85 per cent within four years

Outcome: on average, users were worse off

Guasch, J. L. (2004), Granting and Renegotiating Infrastructure Concessions: Doing It Right, World Bank Institute, Tables 1.7 to 1.9 (pages 13–14) and page 12.

Contract Features and Renegotiation



Award criterion: 60 per cent renegotiated when awarded on the lowest tariff, 11 on the highest transfer fee

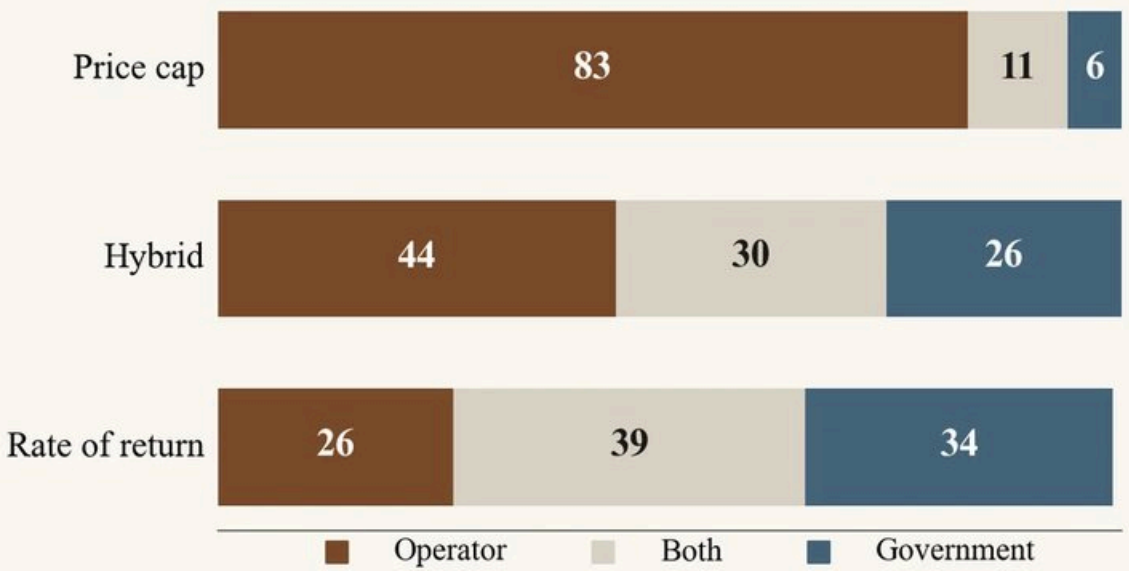
Obligations: 70 per cent with investment requirements, 18 with performance indicators

Process: 46 per cent when bid competitively, 8 when negotiated bilaterally

Reading: competition for the market buys a low price and a contract that is later reopened

Guasch, J. L. (2004), Granting and Renegotiating Infrastructure Concessions: Doing It Right, World Bank Institute, Table 1.16 (page 17) and Table 1.11 (page 15).

Initiators of Renegotiation



Who asks: operators requested 61 per cent of renegotiations and governments 26

Price caps: operators initiated 83 per cent of renegotiations

Rate of return: governments initiated 34 per cent, and operators 26

Design: a guarantee and cap agreed in advance replace a negotiation after demand falls short

Guasch, J. L. (2004), Granting and Renegotiating Infrastructure Concessions: Doing It Right, World Bank Institute, Tables 1.13 and 1.14 (page 16); Engel, Fischer and Galetovic (2013), page 2.

Case Study: Heathrow Airport Expansion (Hannah Newell, DfT)

Planning: a nationally significant infrastructure project

The state and its private partners: how the relationship works

Stakeholders: the governance of a contested project

Discussion: questions for the guest

Summary

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